

A Bold Amateur Approach to the Symmetry Underlying the Riemann Hypothesis

Mohamed Amine Belachhab - Independent HS researcher

27/05/2025

Preface

And just as a disclaimer, the try is completely innovative and bold, so please before you start judging please change your lenses ;). Thanks in advance my dears!

Also note: I AM NOT A PROFESSIONAL MATHEMATICIAN AND THAT THIS IS ONLY AMATEUR TEENAGE WORK DONE WITH VERY LITTLE RESSOURCES, anyway sending LOVE!

Abstract

In this very modest paper I will try to somewhat give an argument supporting the Hypothesis all while adding a little spicy theory... Stay tuned ;)

Introduction

No ideas.

Try

We know that

$$\zeta(s) = f(s) \cdot \zeta(1-s)$$

So if s is a zero of the Zeta function then $1-s$ is also one, however, this rule only applies to non-trivial solutions (complex ones) because trivial ones make the $f(s)$ part crush into a 0. So, some kind of absolute symmetry is there, and which has been rigorously proven by fellow mathematicians!

Oh! And let's not forget that complex solutions come in pairs (conjugates): $a + bi$ and $a - bi$, and almost never if not never in quadruples.

And we also know that each one of those 2 complex solutions behaves in the same way as the other one.

And just one additional thing, let s be a complex solution of the Zeta function, so according to the symmetry, $1 - s$ is also a complex solution, which means that

$$\begin{cases} s = a + bi \\ 1 - s = 1 - a - bi \end{cases}$$

we observe the presence of one solution of our pair of solutions ($s = a + bi$), however the other one is absent from which we conclude that the other one ($1 - s = 1 - a - bi$) is just the conjugate of the first solution, so

$$\bar{s} = 1 - a - bi = a - bi$$

from which

$$1 - a = a$$

which is true only and only if $a = \frac{1}{2}$, another argument; for the symmetry to absolutely hold true and for the function to respect the well-known pattern of math: "Complex solutions come in conjugates; pairs."

Contradiction Argument

Let's suppose that a solution exists with $a \neq \frac{1}{2}$

The solutions become: $(a + bi; a - bi; 1 - a - bi; 1 - a + bi)$

And these 4 solutions give a sense of **UNCERTAINTY** in the behaviour of the Zeta function which is unacceptable due to the **ABSOLUTE SYMMETRY FORCED BY THE NATURE OF THE ZETA FUNCTION**.

And to push away the "Symmetry isn't enough to prove that all non-trivial solutions lie on the critical line $a = \frac{1}{2}$," I present to you my theory as the conclusion.

Conclusion

"Actually, maybe the symmetry itself is a result of the placement of the non-trivial solutions, that appears in other forms that made proving the symmetry possible through other routes, and even that maybe we can't prove or disprove the hypothesis except by going the opposite

way: from the symmetry to the placement of the solutions.”

Here you go world, maybe the problem wasn't our incompetence to prove or disprove this hypothesis, maybe it was our incompetence to choose the way to do it ;).

END